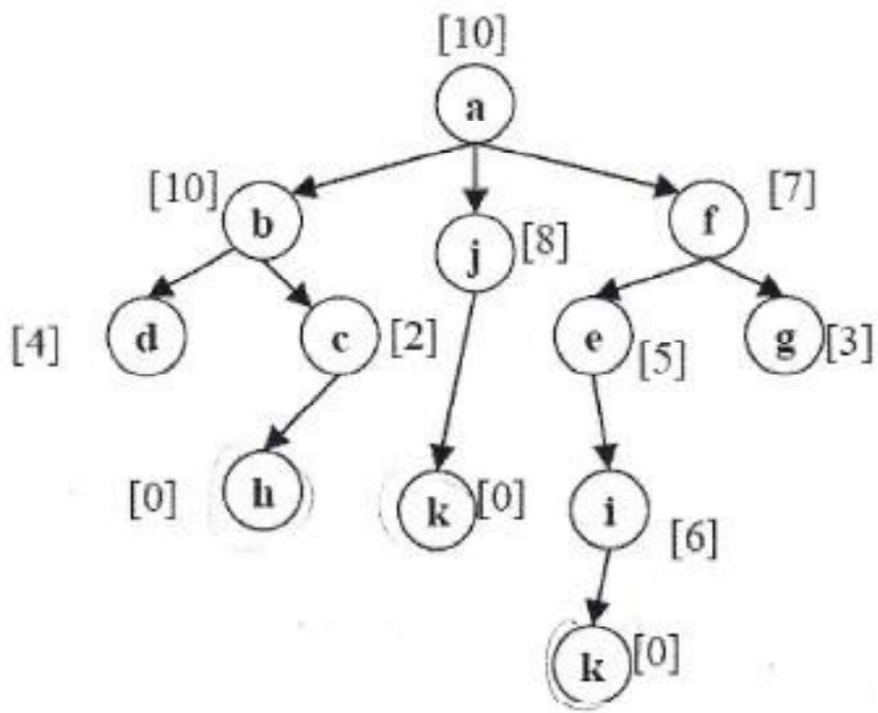
Beam Search

Hill Climbing Vs. Beam Search

- Hill climbing just explores all nodes in one branch until goal found or not being able to explore more nodes.
- Beam search explores more than one path together. A factor **k** is used to determine the number of branches explored at a time.
- If k=2, then two branches are explored at a time. For k=4, four branches are explored simultaneously.
- The branches selected are the **best branches** based on the used **heuristic evaluation function**.

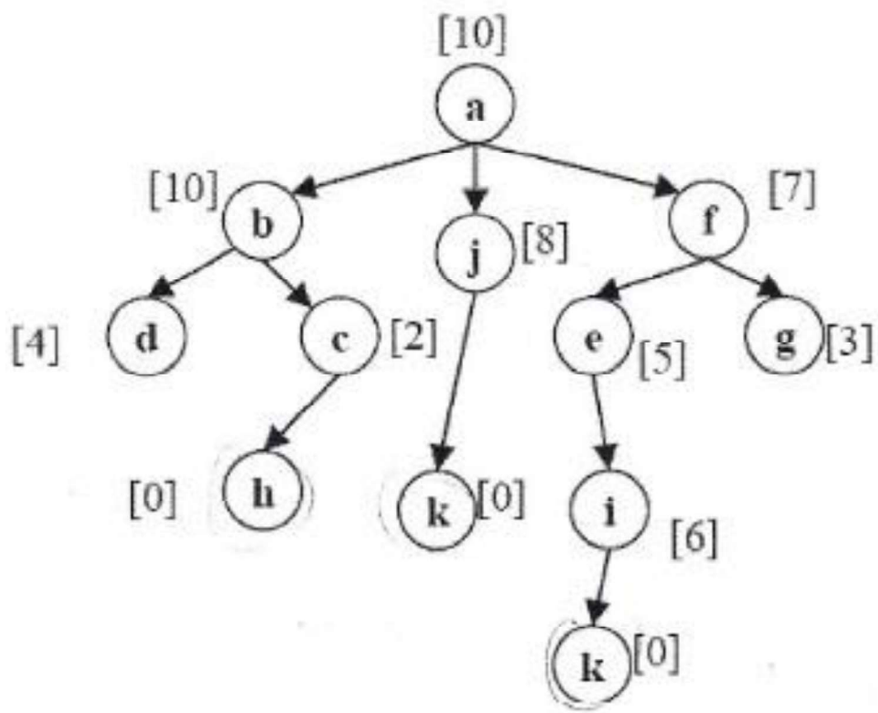
Beam Search, k=2
Goal – **Node K**

Current	Children
a	---

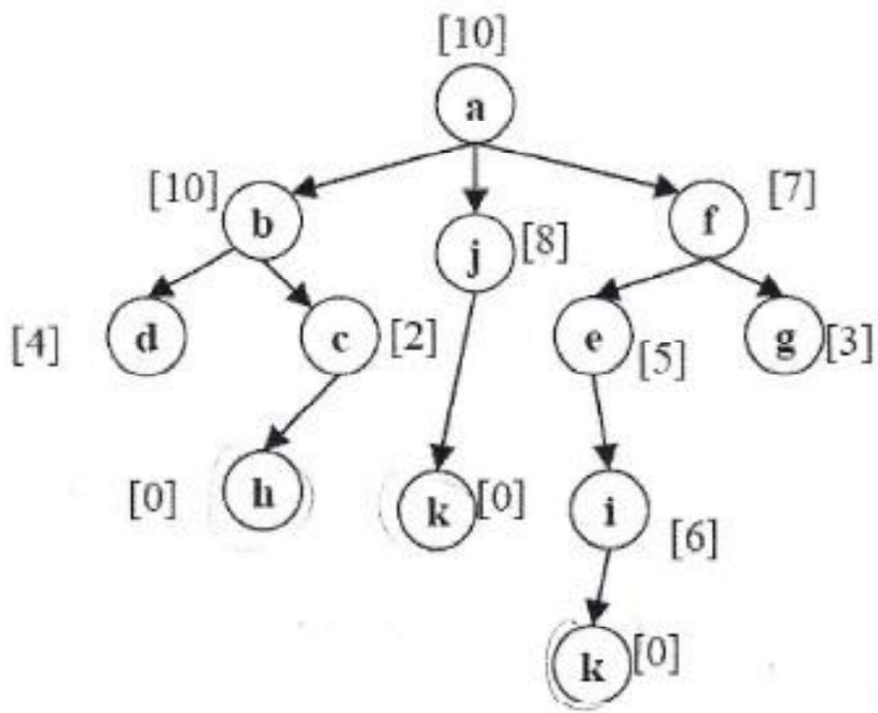


Beam Search
Goal – **Node K**

Current	Children
a	---

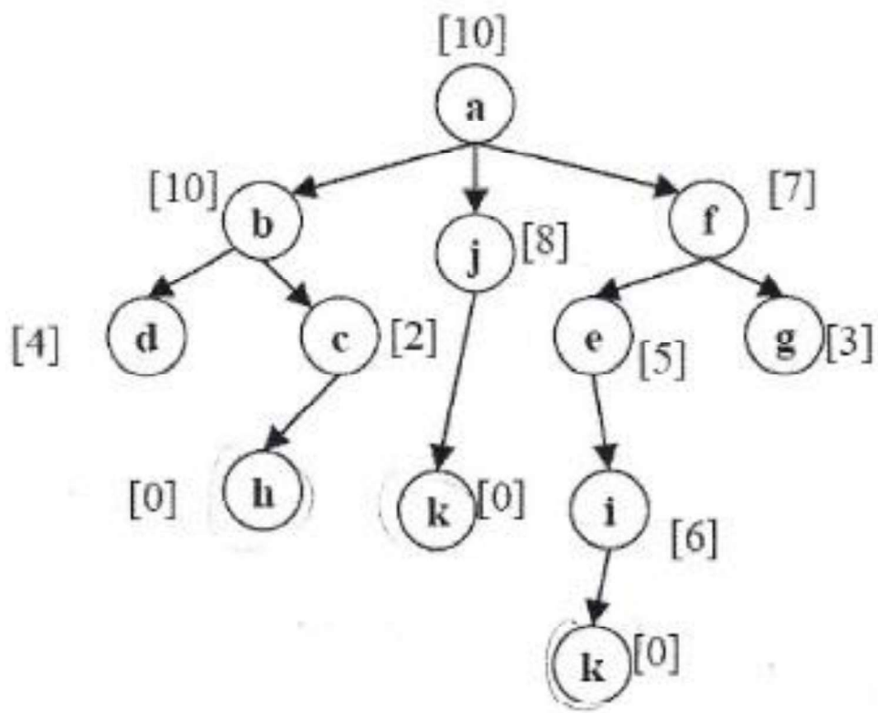


Beam Search
Goal – **Node K**



Current	Children
a	---
a	

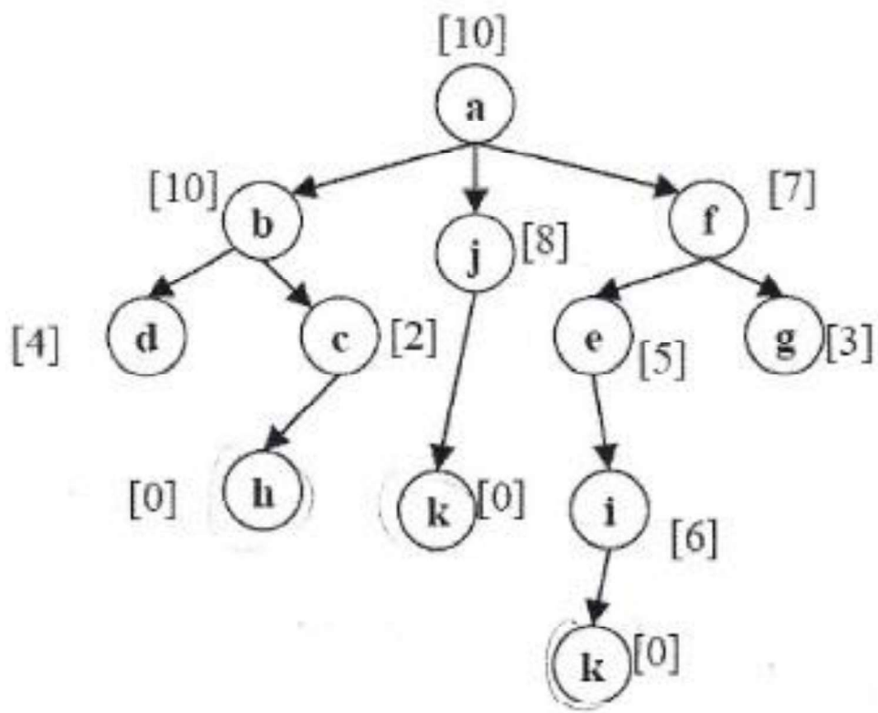
Beam Search
Goal – **Node K**



Current	Children
a	---
a	f_7, j_8, b_{10}

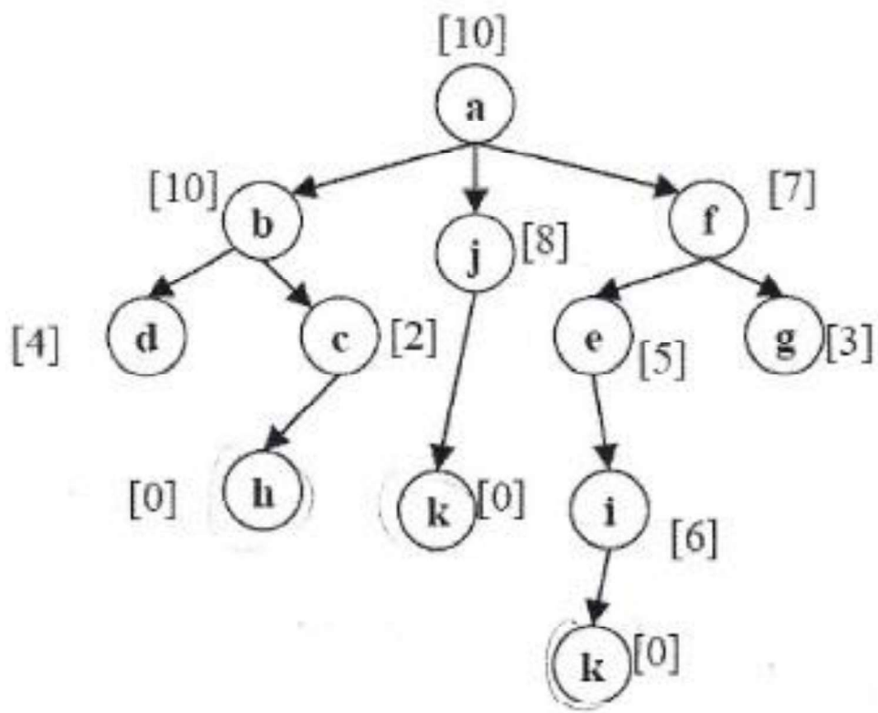
Beam Search
Goal – **Node K**

Best k Successors	Current	Children
	a	---
	a	f_7, j_8, b_{10}

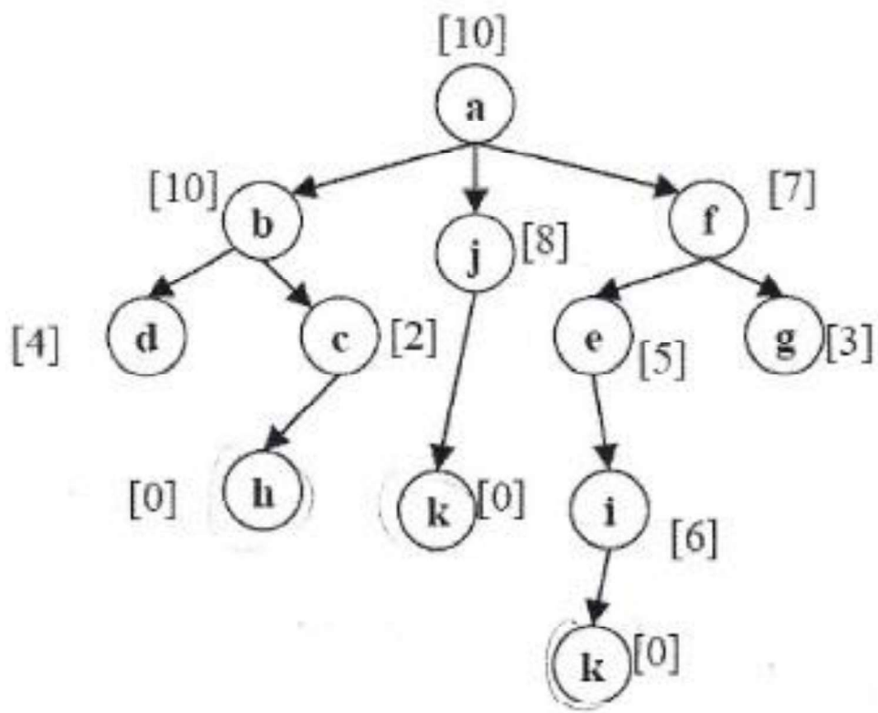


Beam Search
Goal – **Node K**

Best k Successors	Current	Children
	a	---
	a	f_7, j_8, b_{10}

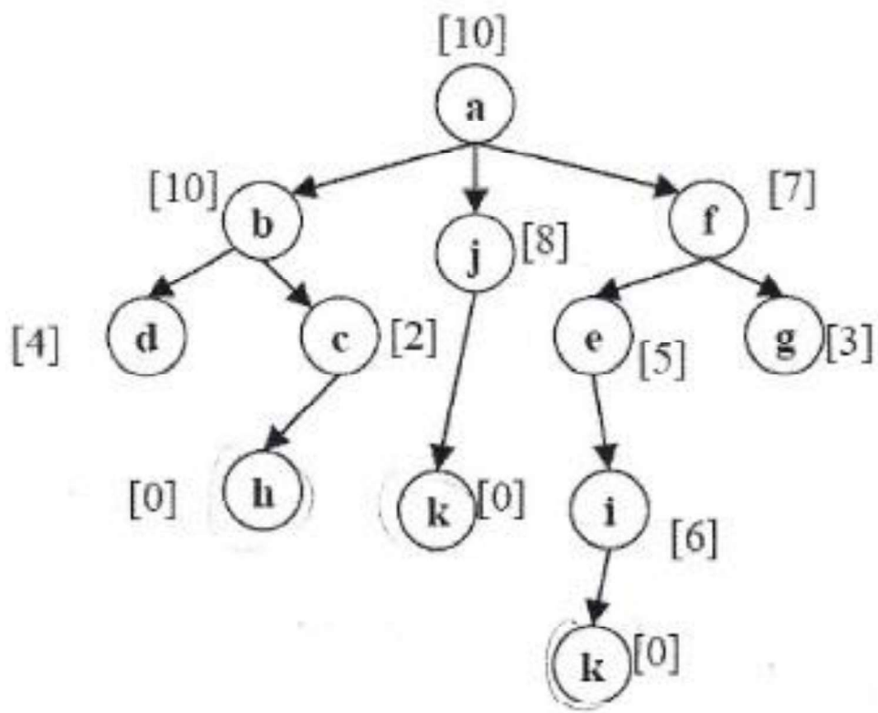


Beam Search
Goal – **Node K**



Current	Children
a	---
a	f_7, j_8, b_{10}
f j	

Beam Search
Goal – **Node K**



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a	---
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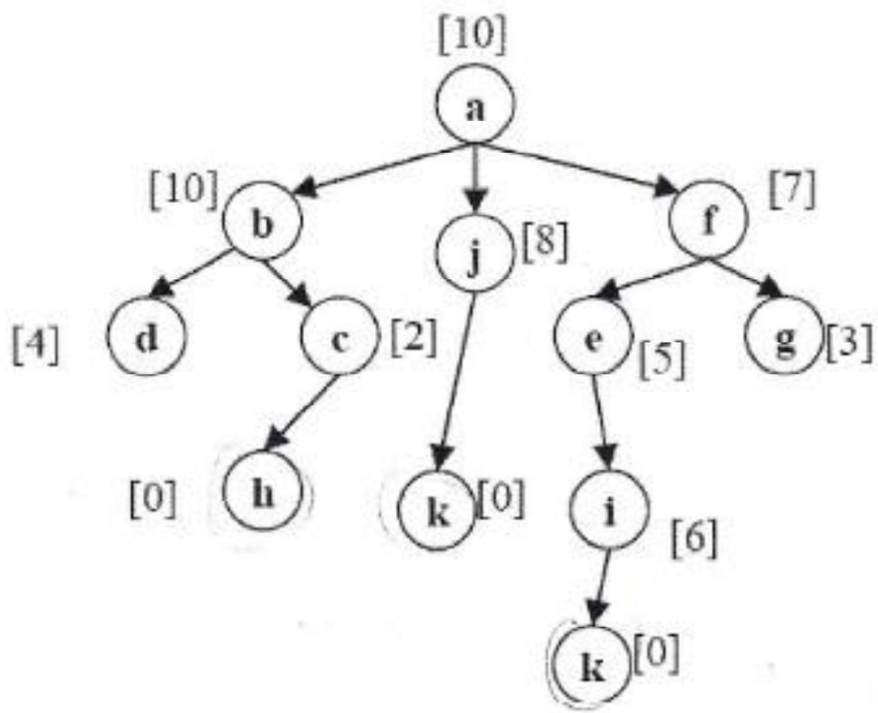
f j	

Beam Search
Goal – **Node K**



Current	Children
a	---
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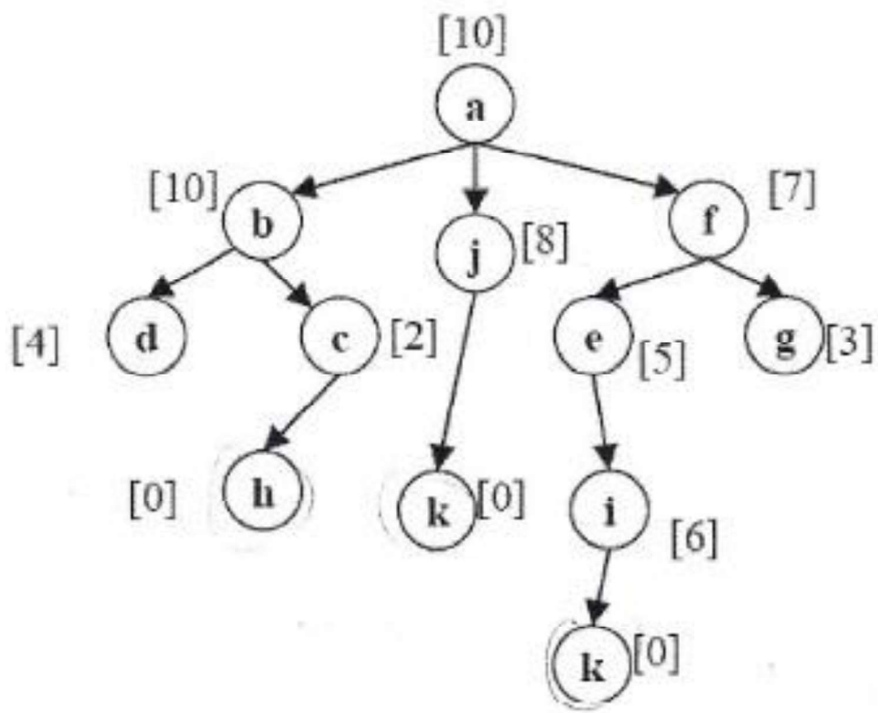
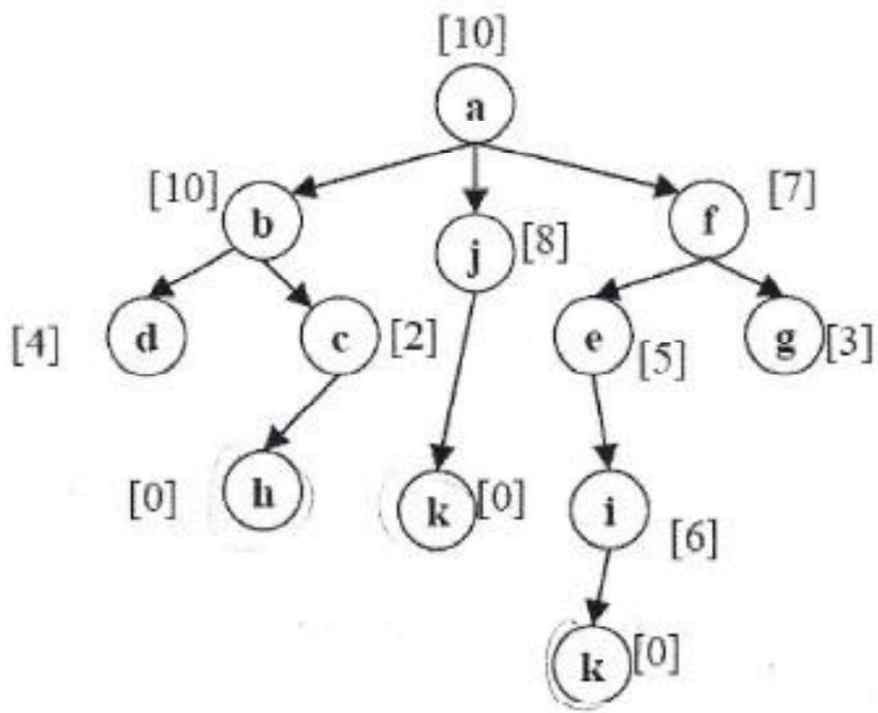
f j	

Beam Search
Goal – **Node K**



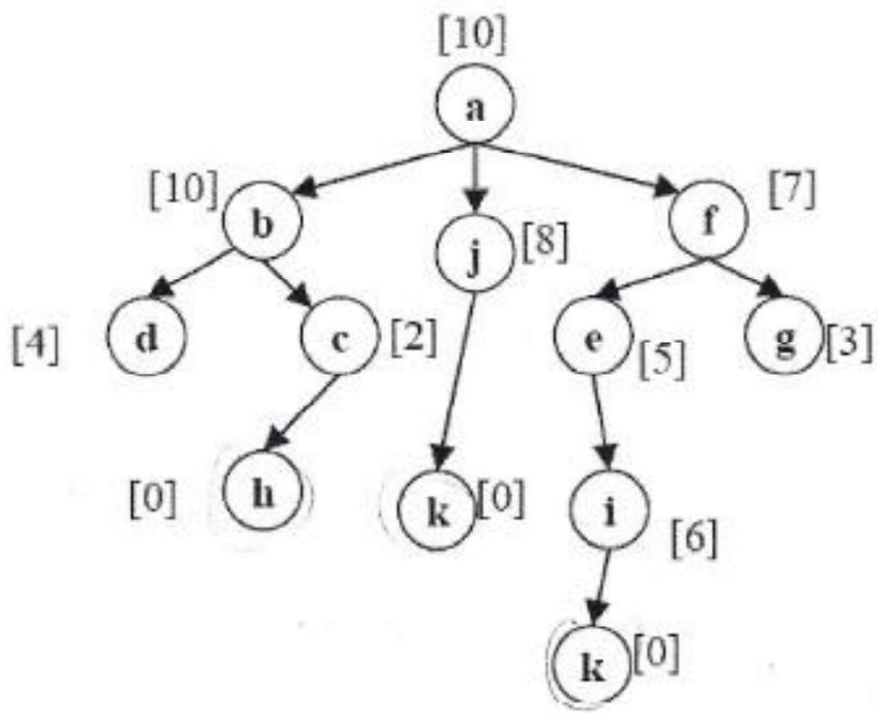
Current	Children
a	---
a	f_7, j_8, b_{10}
f	j
g_3, e_5	k_0

Beam Search
Goal – **Node K**



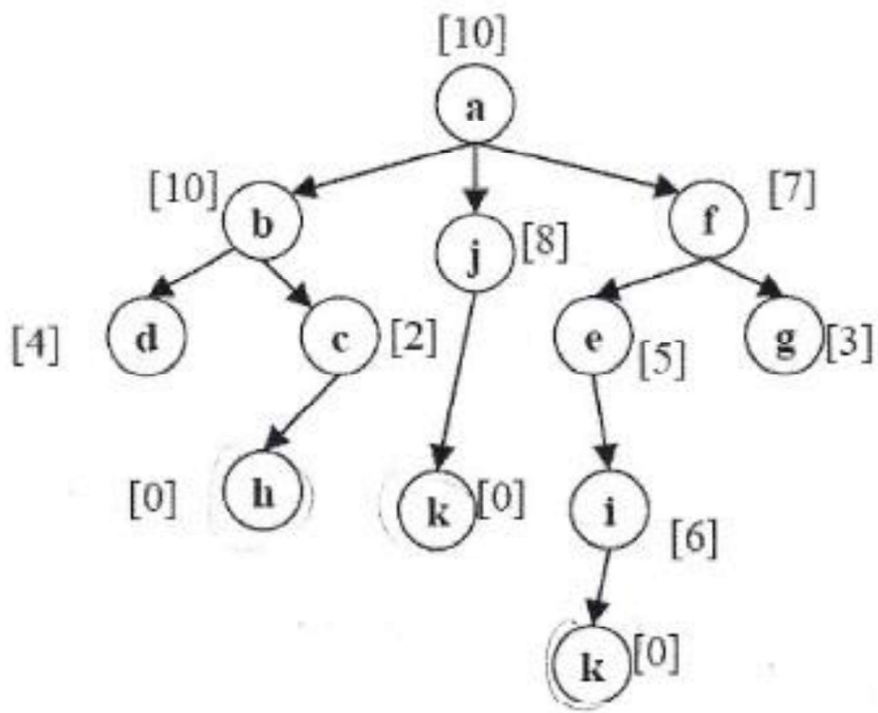
Current	Children
a	---
a	f_7, j_8, b_{10}
f j	
g_3, e_5 k_0	
k_0, g_3, e_5	

Beam Search
Goal – **Node K**



Best k Successors	Current	Children
	a	---
	a	f_7, j_8, b_{10}
	f j	
	g_3, e_5 k_0	
	k_0, g_3, e_5	

Beam Search
Goal – **Node K**



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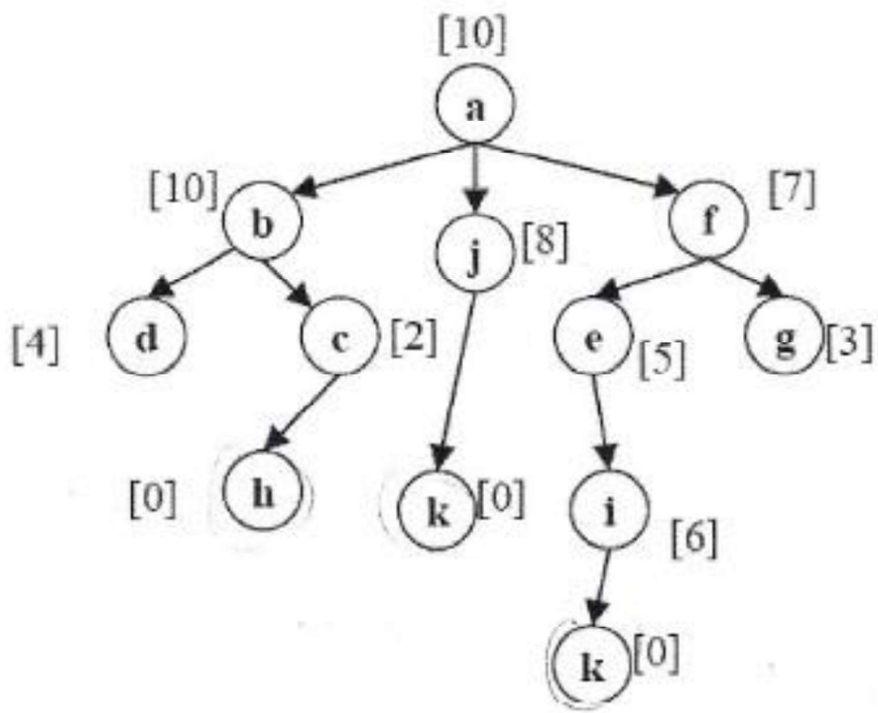
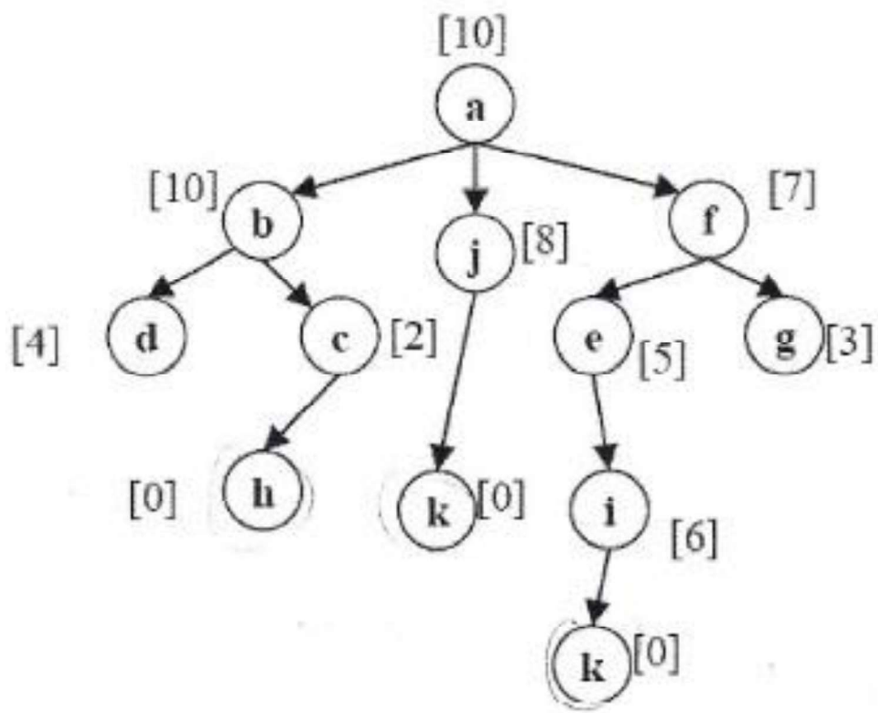
Beam Search
Goal – **Node K**



Current	Children
a	---
a	f_7, j_8, b_{10}
f j	
g_3, e_5 k_0	
k_0, g_3, e_5	
k g	

Beam Search
Goal – **Node K**



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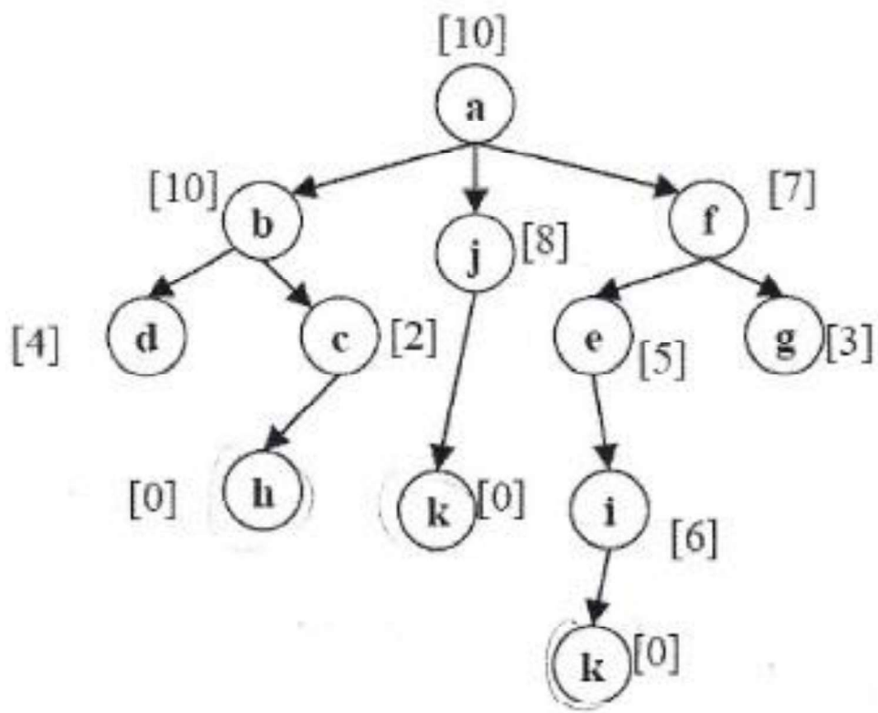
g_3, e_5 k_0	
k_0, g_3, e_5	
k g	

Beam Search
Goal – **Node K**



Current	Children
a	---
a	f_7, j_8, b_{10}
f j	
g_3, e_5 k_0	
k_0, g_3, e_5	
k g	
GOAL	